

1. INTRODUCTION

Python has rapidly emerged as one of the most dominant and versatile programming languages in the modern technological era. Its clean syntax, vast ecosystem of libraries, and supportive community have made it a go-to language in numerous fields, especially in data science and artificial intelligence (AI). From web development to automation, and from machine learning to deep learning, Python serves as the backbone for a multitude of innovative solutions.

Artificial Intelligence (AI), on the other hand, has shifted from being a buzzword to becoming the heart of cutting-edge technology across industries. AI refers to the simulation of human intelligence by machines—enabling them to perform tasks such as reasoning, learning, and decision-making.

Within AI, subfields such as machine learning (ML) and deep learning (DL) have gained massive traction. These approaches allow computers to learn from data and improve over time without being explicitly programmed for every single task.

In recent years, the integration of AI into video surveillance systems has opened new avenues in public safety, law enforcement, and emergency response. One of the most promising applications is **automated crime scene detection**—the ability of a system to analyze live or recorded video footage and detect potentially criminal activities in real time. This not only assists authorities in faster decision-making but also reduces manual monitoring fatigue and errors.

Python, renowned for its simplicity and extensive scientific libraries, serves as the foundation of this project. Leveraging powerful frameworks like TensorFlow and Keras, the research builds and trains a customized CNN model based on the EfficientNetB0 architecture. This model not only classifies images as "real" or "fake" but also optimizes accuracy through image preprocessing techniques, activation functions, pooling layers, and dropout strategies to combat overfitting.

Python's robust ecosystem enables seamless integration of deep learning pipelines, data augmentation via ImageDataGenerator, and real-time image analysis with OpenCV, making the model deployment both efficient and scalable.

This project, titled **"Crime Scene Detection Using Deep Learning,"** leverages Python-based deep learning frameworks to analyze short video clips and categorize them into one of four crime-related activities: **Arrest, Abuse, Arson, and Assault**.

The solution utilizes 3D Convolutional Neural Networks (3D CNNs), which are capable of capturing both spatial and temporal features from video sequences. The model is trained on labeled video datasets and integrated into a user-friendly interface using **Streamlit**, allowing users to upload video clips and instantly view the system's prediction.

The goal of this project is to contribute a fast, intelligent, and scalable crime detection mechanism that could eventually be deployed in real-world scenarios like surveillance cameras, smart city infrastructure, or security systems.

1.1 MACHINE LEARNING

Machine learning plays a critical role in crime scene detection by enabling AI models to generate meaningful textual descriptions for images. Deep learning, a subset of machine learning, has significantly improved the accuracy of crime scene detection systems by utilizing neural networks to extract complex features from images and generate human-like text. Convolutional neural networks (CNNs) are commonly used for feature extraction, as they excel at identifying objects, textures, and spatial relationships within images. These extracted features are then processed by recurrent neural networks (RNNs) or transformer-based models to generate captions in natural language.

The importance of machine learning extends far beyond crime scene detection. In healthcare, ML models are used for diagnosing diseases, analyzing medical images, and predicting patient outcomes. In finance, machine learning helps detect fraudulent transactions, optimize trading strategies, and provide personalized recommendations. In the automotive industry, ML powers self-driving cars by enabling real-time decision-making based on sensor inputs. Additionally, machine learning is widely used in speech recognition, recommendation systems, chatbots, and many other applications that impact daily life.

As machine learning continues to evolve, researchers are exploring ways to make models more efficient, interpretable, and ethically responsible. Challenges such as bias in training data, high computational costs, and the need for large datasets are being addressed through techniques like transfer learning, model pruning, and federated learning.

The future of machine learning is expected to bring even more sophisticated models capable of understanding and generating human-like language, improving real-world applications such as crime scene detection, conversational AI, and multimodal learning systems.

1.2 DEEP LEARNING

Deep learning has played a crucial role in enhancing crime scene detection systems by enabling AI models to generate highly accurate and contextually relevant descriptions of images. The combination of CNNs for feature extraction and RNNs or transformers for text generation has significantly improved the quality of automated captions. Moreover, attention mechanisms have further refined scanning models by allowing them to focus on specific regions of an image while generating descriptions. These improvements have made deep learning-powered crime scene detection systems highly effective for applications such as assistive technologies, content recommendation, and digital accessibility.

As deep learning continues to evolve, research is focused on making models more efficient and interpretable. Techniques like transfer learning, model compression, and federated learning are being explored to reduce the computational demands of deep learning models while maintaining high accuracy. Additionally, ethical concerns such as bias in AI models and data privacy are being actively addressed to ensure responsible AI development. The future of deep learning holds immense potential, with ongoing advancements expected to further improve AI-driven applications, including crime scene detection, speech synthesis, and autonomous decision-making systems.

For sequential data processing, Recurrent Neural Networks (RNNs) are widely used. Unlike traditional feedforward networks, RNNs have a feedback loop that allows them to retain information from previous steps, making them suitable for tasks like speech recognition, time-series forecasting, and language modeling. However, standard RNNs suffer from issues such as vanishing gradients, which limit their ability to capture long-term dependencies. To address this, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) were introduced, enabling more effective learning of long-range dependencies.

1.3 OBJECTIVE'S

With the increasing demand for intelligent surveillance systems, the use of deep learning to detect and categorize crime scenes in real-time video feeds has emerged as a crucial advancement in public safety and forensic analysis. This project aims to develop a lightweight yet powerful AI-driven solution capable of recognizing criminal activity across four categories—**Arrest, Abuse, Arson, and Assault**—by processing video inputs using 3D Convolutional Neural Networks (3D CNNs). Built using Python, PyTorch, and Streamlit, the system offers an interactive interface and is optimized for efficient deployment.

- **The primary objective** of this project is to create a robust deep learning model capable of accurately classifying short video clips into predefined crime-related categories. By leveraging the temporal and spatial feature extraction power of 3D CNNs, the system captures context from motion and frames, providing reliable detection across varied input samples.
- **Another key goal** is to build an effective video preprocessing pipeline. This includes frame extraction, resizing, normalization, and conversion into fixed-length clips, ensuring consistency and minimizing data irregularities. Proper preprocessing is vital for enhancing model performance and generalization to real-world inputs.
- **An important objective** is to deliver an accessible and user-friendly application interface. With Streamlit, users can simply upload a video clip and receive instant classification feedback, eliminating the need for technical expertise and bridging the usability gap between complex AI systems and end-users like law enforcement personnel.

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- **A significant focus** is on optimizing the model for both performance and efficiency. By fine-tuning only essential layers and incorporating dropout for regularization, the model strikes a balance between accuracy and speed—making it viable for real-time or near-real-time deployment, even on modest hardware setups.
 - **To ensure the system's reliability**, detailed performance evaluation is integrated using standard metrics such as accuracy, confusion matrix, and loss visualization. These tools help in monitoring training behavior, identifying overfitting or underfitting, and refining the model iteratively.
 - **Finally**, the project is designed with scalability in mind. The modular codebase allows for easy integration of new crime categories, real-time video feed support, or model upgrades like Transformers or hybrid CNN-RNN architectures, ensuring the system can evolve as surveillance needs and AI capabilities expand.

The overarching goal is to build an intelligent, real-time crime detection solution that can assist in early incident identification, improve response times, and ultimately contribute to safer environments through AI-powered automation.

1.4 PROBLEM STATEMENT

In an era where surveillance systems, digital forensics, and video documentation have become integral to public safety and criminal investigations, the ability to analyze and interpret visual content accurately has never been more critical. However, despite the massive influx of CCTV footage and video evidence available for crime detection, manual review remains the dominant approach—leading to significant delays, errors, and inefficiencies. Human monitoring is not only time-consuming but also susceptible to fatigue, oversight, and subjective bias, particularly in high-stress or high-volume situations.

Traditional computer vision techniques have provided some automation in object and motion detection but fall short in understanding complex, dynamic scenes like crimes in progress. These conventional algorithms often lack the temporal awareness and contextual understanding necessary to differentiate between violent or suspicious activity (such as **arson**, **assault**, **abuse**, or **arrest**) and benign movements. This gap has led to underutilized surveillance resources and missed opportunities for timely intervention or evidence gathering.

Moreover, with crime rates continuing to rise in densely populated and urban areas, the volume of video content requiring analysis has reached unmanageable scales. Law enforcement agencies and security firms struggle to process this data in real-time, leading to delayed responses and compromised public safety. In many cases, video footage is reviewed only after an incident has occurred—limiting its use for prevention or active intervention.

The challenge becomes even more pronounced when trying to build scalable and accessible crime detection systems. Many existing AI solutions demand high-end GPUs, massive storage capacities, and advanced technical knowledge, making them inaccessible to mid-level organizations or rural deployments. On the other hand, lightweight models often sacrifice accuracy, failing to detect nuanced or subtle patterns indicative of crime.

In addition, general-purpose action recognition models are not optimized for crime-specific datasets. They may misclassify critical events due to insufficient domain training, imbalanced datasets, or poor temporal modeling. Most systems also lack intuitive user interfaces, making it hard for law enforcement officers or non-technical staff to interact with them efficiently.

This project tackles these critical limitations by proposing a **deep learning-based video classification system** specifically tailored for **crime scene detection**. Using a **3D Convolutional Neural Network (3D CNN)** trained on short video clips labeled across four distinct crime types—**Arrest, Abuse, Arson, and Assault**—the model captures both spatial and temporal cues to identify criminal activity with high precision. The system also integrates a **Streamlit-based interactive web interface**, enabling easy video upload, live classification, and immediate feedback, all without requiring technical expertise.

Key challenges addressed by this project include:

- The need for **automated, real-time video analysis** capable of detecting and categorizing criminal activities.
- The lack of **efficient, scalable, and resource-friendly models** deployable on standard systems without performance degradation.
- The demand for a **crime-specific, domain-trained AI model** rather than generic action recognition solutions.
- The importance of **user accessibility and interface simplicity**, ensuring the technology can be widely adopted by law enforcement and non-technical users.
- The necessity to **reduce latency** and enhance **detection speed and accuracy**, especially for time-sensitive surveillance applications.

Ultimately, this project aims to bridge the gap between complex deep learning models and real-world law enforcement needs, empowering security agencies with a tool that not only enhances crime detection but also lays the foundation for proactive surveillance, quicker response, and more reliable public safety systems.

2. LITERATURE SURVEY

Crime scene detection has become an increasingly critical area of research due to the growing demand for intelligent surveillance and public safety systems. As urban environments become more reliant on camera networks and video footage for security, there is an urgent need for systems that can automatically and accurately analyze video content to identify criminal activity in real-time. With the advent of deep learning, particularly spatiotemporal models like 3D Convolutional Neural Networks (3D CNNs), researchers have made significant progress in developing automated crime detection systems that learn complex motion patterns and spatial context directly from video sequences. This section provides an overview of prominent studies and contributions relevant to crime scene analysis, deep learning in surveillance, and video classification.

1. Patil, A., et al. (2021) – Video Surveillance using Deep Learning for Crime Detection

- This study introduces a deep learning-based surveillance system that classifies live video feeds for crime activity using Convolutional Neural Networks.
 - The researchers propose a modular pipeline combining object detection, motion tracking, and activity classification. While the model achieves decent accuracy on staged crime datasets, its reliance on 2D CNNs limits its ability to fully grasp temporal features present in video streams.
 - The study emphasizes the need for spatiotemporal models to improve context understanding.
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2. Kar, A., et al. (2022) – Human Action Recognition using 3D Convolutional Neural Networks

- Kar and colleagues explore the application of 3D CNNs in action recognition—a core component in crime scene classification. Their work demonstrates that 3D CNNs outperform 2D CNNs by learning both spatial and temporal features simultaneously from short video clips.
 - Using datasets like UCF101 and HMDB51, the model achieves high accuracy in identifying violent actions, laying a foundation for their application in crime detection.
 - Their findings validate the effectiveness of clip-based training for real-world video classification.
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3. Li, X., et al. (2020) – Crime Detection from Surveillance Videos using Deep Neural Networks

- This paper presents a hybrid approach combining LSTM layers with CNNs to capture temporal sequences of human actions.
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- The study utilizes violent vs. non-violent binary classification and applies the model to public datasets like CCTV-Fights and Violent-Flows. Though the method shows strong potential, real-time implementation remains challenging due to computational complexity.
 - It further underlines the importance of lightweight yet robust models for deployment in resource-limited environments.
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4. Sultani, W., et al. (2018) – Real-World Anomaly Detection in Surveillance Videos

- Sultani et al. introduce an unsupervised deep learning framework for anomaly detection using video surveillance. Their system doesn't rely on predefined crime categories but instead identifies statistically abnormal behaviors using autoencoders and MIL (Multiple Instance Learning).
 - Although this method is effective for general anomaly detection, it lacks the specificity required for classification into detailed crime types such as arson or assault. This research, however, paves the way for incorporating anomaly scoring into supervised crime classification tasks.
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5. Tran, D., et al. (2015) – Learning Spatiotemporal Features with 3D Convolutional Networks

- This landmark paper introduces the C3D architecture—a 3D CNN model trained on video datasets to extract deep features for action recognition. It significantly influenced later work on spatiotemporal modeling by demonstrating that 3D kernels are well-suited for learning motion dynamics in videos.
 - While not specifically targeted at crime detection, C3D's architecture has inspired applications in surveillance, behavior analysis, and security systems. The study provides critical insight into how crime classification systems can be trained on temporal slices of video.
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6. Vishwakarma, S., et al. (2019) – Survey on Crime Scene Detection and Prediction using AI

- In this survey, the authors outline various approaches for AI-based crime detection, categorizing them into prediction, classification, and anomaly detection models. They discuss challenges such as data imbalance, ethical concerns, and real-time inference constraints.
 - The paper emphasizes the growing role of deep learning models—especially CNNs and hybrid LSTM-CNNs—in transforming surveillance analytics. The authors advocate for crime-specific datasets and suggest the integration of models into real-world, user-accessible interfaces.
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7. Zhao, C., et al. (2021) – Efficient Deep Learning Models for Mobile Video Surveillance

- Focusing on resource efficiency, this study proposes using lightweight architectures like MobileNet and EfficientNet for video analytics on mobile platforms. Though primarily applied to person detection and face recognition, their findings are relevant to crime scene detection systems aiming for broader deployment.
 - The paper validates that optimized deep learning models can retain accuracy while dramatically reducing computation cost—an approach relevant to this project's use of short video clips and lightweight deployment through Streamlit.
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The body of literature around crime scene detection highlights an ongoing shift from rule-based surveillance systems to data-driven, AI-enabled solutions. While traditional approaches relied heavily on handcrafted features or motion tracking, modern systems increasingly leverage deep learning models—especially 3D CNNs—for their ability to learn complex patterns across both space and time. The reviewed studies underscore key considerations such as real-time performance, model efficiency, dataset diversity, and interface accessibility.

Despite promising results, challenges remain in deploying these systems across diverse environments, managing class imbalance in crime datasets, and maintaining accuracy under occlusion and poor lighting.

This project builds upon these findings by creating a modular, scalable, and interactive crime detection system using 3D CNNs and Streamlit—offering a blend of precision, speed, and usability that aligns with current research trends while addressing real-world deployment constraints.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Currently, crime scene detection systems primarily leverage deep learning models—most notably **Convolutional Neural Networks (CNNs)**, **Recurrent Neural Networks (RNNs)**, and hybrid architectures—for analyzing surveillance footage and identifying potentially criminal activities. These models are trained to recognize specific human actions (e.g., fighting, running, or fire-related activity) using frame-level visual cues and temporal motion patterns.

Many systems utilize **2D CNNs** for frame-by-frame image classification, while others integrate **Long Short-Term Memory (LSTM)** networks to capture sequential dependencies in video clips. Recent approaches have incorporated **3D CNNs**, which extract both spatial and temporal features from short video sequences, offering improved accuracy for complex action recognition.

Models are often trained on action recognition datasets such as **UCF Crime**, **Avenue Dataset**, **CCTV-Fights**, and **Violence Flow**, with crime detection defined as a classification task (e.g., “violent” vs. “non-violent” or “normal” vs. “abnormal”). Systems are sometimes deployed in smart surveillance environments to flag suspicious activities for human review, but they still fall short of robust, multi-class crime classification in real-world scenarios.

However, existing systems face several limitations:

1. **Inconsistent Real-Time Performance:**

Many current systems lack the computational efficiency to process continuous video feeds in real-time, especially when using high-resolution footage. This results in delayed detections or system bottlenecks, which are unacceptable in time-sensitive security applications.

2. **Limited Action Categories:**

Most existing models are restricted to binary or low-classification tasks (e.g., “violent” vs. “non-violent”) and are not designed to distinguish between different types of crime (e.g., **arrest**, **arson**, **assault**, **abuse**). This hampers the usefulness of such systems in legal or forensic investigations.

3. **Dataset Dependency and Lack of Generalization:**

Current models are often trained on domain-specific datasets and perform poorly when tested on videos from different sources or environments (e.g., new camera angles, lighting conditions, or unseen locations). Generalizing across real-world data remains a challenge.

4. **Neglect of Multi-Modal Features:**

Most detection systems rely solely on visual features, ignoring **audio**, **scene context**, or **temporal metadata**—all of which can significantly enhance model accuracy.

For instance, a scream or explosion sound could be a strong indicator of a crime event, but these are often overlooked.

5. **Difficulty Handling Subtle or Non-Obvious Crimes:**

Crimes like emotional abuse, theft without force, or coordinated illegal activity can be visually subtle and context-dependent, making them hard to classify with traditional video-only models. Current systems often fail to capture these nuanced scenarios.

3.2 PROPOSED SYSTEM

To address the limitations of existing crime scene detection systems, this project proposes a robust and scalable deep learning-based solution capable of detecting and classifying various crime types—**arrest**, **abuse**, **arson**, and **assault**—from surveillance or recorded video footage. The system leverages a **3D Convolutional Neural Network (3D CNN)** architecture trained on short video segments to capture both **spatial** and **temporal features**, enabling accurate identification of dynamic criminal activities.

The primary design goal is to create a lightweight, efficient, and interactive application that delivers high accuracy without demanding high-end computational infrastructure. This makes the system deployable in real-world environments, including **low-resource settings like small security centers, schools, or public transport stations**.

Key Components of the Proposed System:

1. **3D CNN-Based Video Classification:**

- The core of the system is a **custom 3D CNN model** trained on **16-frame video clips**. Unlike 2D CNNs that treat each frame independently, the 3D CNN extracts **spatiotemporal patterns**—such as motion trajectories, object interactions, and violent behavior transitions.
- This architecture enables the system to understand the **context of movement**, essential for differentiating between types of crimes that may appear visually similar in isolated frames.

2. **Custom Crime Video Dataset:**

- The model is trained on a custom-labeled dataset structured by folders for each crime class: **Arrest**, **Abuse**, **Arson**, and **Assault**, with all videos preprocessed into 16-frame clips.
- A **‘background’ class** is used to represent non-crime or neutral video content, helping the model reduce false positives.

3. **Efficient Preprocessing Pipeline:**

- Each video is downsampled into fixed-size clips, normalized, and resized for uniformity across the dataset.
 - Augmentation techniques such as frame flipping, rotation, and brightness shifts are applied to improve model generalization on unseen data.
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4. Streamlit Web Interface:

- An **interactive web app** built using Streamlit allows users to **upload video files**, which are automatically sliced into clips, fed to the model, and classified.
- The output includes the **predicted crime category**, **confidence score**, and visual feedback, enabling fast human review and actionable insights.

5. Optimized for Mid-Range Deployment:

- The model is trained with performance in mind—using early stopping, dropout layers, and batch normalization to ensure smooth inference without overfitting.
- The system is **lightweight enough to run on standard consumer hardware**, making it accessible for local law enforcement or campus security setups.

6. Evaluation & Feedback Loop:

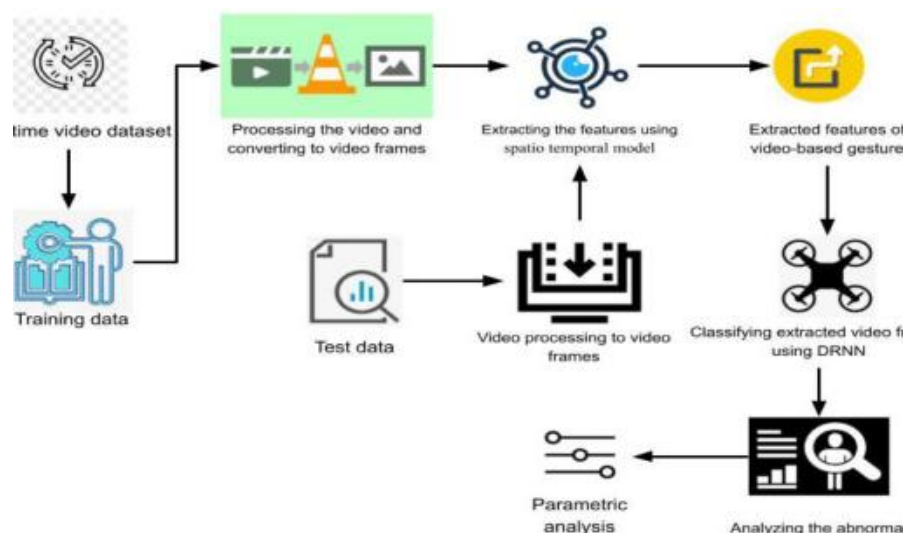
- The system tracks **accuracy**, **precision**, **recall**, and **F1 score** across all classes. Misclassified instances can be manually reviewed and added to a retraining pipeline for continuous learning.
- Real-world usability is improved through a feedback loop where users can mark incorrect predictions and contribute to model fine-tuning.

7. Future-Proof and Modular:

- The modular codebase allows easy expansion to additional crime classes, real-time video stream analysis, or even hybrid models that include **audio cues** and **scene metadata**.
 - With minor changes, the system can be scaled for use in **mobile apps**, **drone surveillance**, or **smart city control rooms**.
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In summary, the proposed system bridges the gap between advanced AI models and real-world security needs. It transforms complex video analysis into an intuitive, high-accuracy tool that not only identifies crimes but also provides the foundation for future enhancements like real-time alerts, multi-modal sensing, and integration into larger security infrastructures.

SYSTEM DESIGN



3.2.1 DEEP LEARNING APPROACH

The deep learning approach in this project revolves around leveraging **3D Convolutional Neural Networks (3D CNNs)** to analyze and classify short video clips into one of four crime categories: **Arrest, Abuse, Arson, or Assault**. Unlike traditional 2D CNNs that analyze static images, 3D CNNs are specifically designed to capture **spatiotemporal patterns**—essential for understanding motion, context, and interactions over time in a crime scene video.

1. Data Collection and Preprocessing

- **Custom Dataset:** A manually curated dataset is used, with each crime class (Arrest, Abuse, Arson, Assault) stored in separate folders. A **‘Background’** class is also included to represent non-criminal activity, helping to minimize false positives during real-world inference.
- **Clip Segmentation:** Each video is sliced into **fixed-length clips** of 16 frames. This allows the model to process consistent temporal chunks, improving its ability to detect motion-based features.
- **Preprocessing Techniques:**
 - **Resizing:** All video frames are resized (e.g., to 112×112 pixels) to reduce computational load and standardize input shape.
 - **Normalization:** Pixel values are normalized to the [0, 1] range to stabilize training.
 - **Augmentation:** Random transformations like flipping, rotation, brightness adjustment, and temporal jittering (frame shuffling) are used to improve generalization.

2. Model Architecture

- **Core Architecture: 3D CNN**
 - The model uses a custom 3D CNN that takes a sequence of 16 consecutive frames as input (with shape [batch_size, 3, 16, 112, 112]) and learns both **appearance** and **motion features**.
 - Layers include 3D convolutions, 3D max-pooling, and ReLU activations, followed by fully connected layers for classification.
- **Network Components:**
 - **Conv3D Layers:** Extract spatial and temporal features simultaneously.
 - **BatchNorm3D:** Normalizes across batches to speed up convergence.
 - **MaxPool3D:** Reduces dimensionality while retaining critical features.
 - **Dropout Layers:** Prevents overfitting by randomly disabling neurons during training.
 - **Fully Connected Layers:**
 - Flattened output is passed through one or more dense layers.
 - Final layer uses **Softmax activation** for multi-class classification (outputting probabilities for the five classes).

3. Model Training

- **Loss Function:** CrossEntropyLoss is used since the problem is a multi-class classification task.
- **Optimizer:** The model is trained using the **Adam optimizer** with a learning rate of 0.0001.
- **Training Parameters:**
 - **Batch Size:** 8 (chosen for balance between memory usage and convergence speed).
 - **Epochs:** Trained for 15–20 epochs with early stopping based on validation loss.
 - **Learning Rate Scheduler:** Reduces learning rate on plateau to refine model performance.
- **Hardware Consideration:** Training is conducted using GPU acceleration (e.g., NVIDIA CUDA), though the final model is optimized to run efficiently on CPU for deployment.

4. Model Evaluation

The model is evaluated on a held-out test set using multiple metrics:

- **Accuracy:** Overall correctness of predictions.
- **Confusion Matrix:** Breakdown of true positives, false positives, etc., per class.
- **Precision & Recall:**
 - **Precision:** How many predicted crimes were actual crimes.
 - **Recall:** How many actual crimes were successfully detected.
- **F1-Score:** Harmonic mean of precision and recall, useful for imbalanced class scenarios.

A detailed classification report is generated after testing, highlighting the strengths and weaknesses of the model on each crime type.

5. Model Deployment (Real-Time Inference)

- The trained model is deployed using a **Streamlit web interface**:
 - **Video Upload:** Users can upload short video clips for analysis.
 - **Frame Extraction:** The uploaded video is split into 16-frame segments.
 - **Inference:** Each clip is passed through the trained model to classify the scene.
 - **Output Display:** The crime type (if any) and confidence score are displayed.
- **Advantages:**
 - Fast prediction time (~1–2 seconds per clip on CPU).
 - No need for external APIs or GPU inference at deployment stage.

6. Performance Enhancement Techniques

To improve performance and robustness:

- **Data Augmentation:** Additional techniques like frame dropout, Gaussian noise injection, and camera motion simulation are planned.
- **Model Checkpointing:** Best models (based on validation F1-score) are saved during training.

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- **Fine-Tuning:** The base 3D CNN is fine-tuned using a smaller, domain-specific dataset to adapt it better for surveillance-style videos.
 - **Ensembling** (*optional*): Predictions from multiple models (e.g., 3D ResNet + custom 3D CNN) can be averaged to improve accuracy.
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In conclusion, this deep learning approach integrates a tailored 3D CNN architecture with effective training, preprocessing, and real-time deployment to offer a practical and powerful tool for crime scene detection. It addresses limitations of prior systems by incorporating temporal dynamics, optimizing for low-resource hardware, and offering an intuitive interface for use in real-world safety monitoring systems.

4. ALGORITHMS

4.1 3D CONVOLUTIONAL NEURAL NETWORKS (3D CNN)

A **3D Convolutional Neural Network (3D CNN)** is an extension of traditional CNNs, designed specifically to process **spatiotemporal data**, such as videos. Instead of analyzing a single image, 3D CNNs process a **sequence of frames** to capture both spatial (what is in the scene) and temporal (how things are moving or changing over time) features. This makes them ideal for **crime scene detection**, where identifying actions like assaults or arrests requires understanding motion, interactions, and context across time.

Working of 3D CNN in Crime Scene Detection

1. Input Video Clip Processing:

- A short clip (e.g., 16 consecutive video frames) is extracted and resized to a fixed size like **112×112 pixels** per frame.
- The clip is structured as a 5D tensor with shape [batch_size, channels, frames, height, width] (e.g., [1, 3, 16, 112, 112] for one RGB video clip).

2. 3D Convolution Operation:

- **3D kernels** slide across the **height, width, and time** dimensions of the clip.
- This allows the model to learn both motion (temporal changes between frames) and visual patterns (edges, objects, interactions).
- For example, it can distinguish between someone walking peacefully vs. being violently pushed (assault).

3. Activation Function (ReLU):

- ReLU introduces non-linearity by zeroing out negative values in the feature maps.
- It enables the network to model complex patterns like violent movement, sudden fire bursts, or physical abuse indicators.

4. 3D Pooling (Downsampling):

- Pooling operations (like **MaxPooling3D**) reduce the size of the 3D feature maps while retaining the most important information.
- This not only reduces computation but also prevents overfitting by abstracting from unnecessary details.

5. Stacked 3D Convolution Layers:

- Deeper layers capture higher-level spatiotemporal patterns, such as:
 - **Body movement patterns** in an arrest.
 - **Flame behavior** in an arson scene.
 - **Physical interactions** in abuse or assault.

6. Flattening and Fully Connected Layers:

- After convolution and pooling, the final 3D feature maps are **flattened into a 1D vector**.
- This vector is passed through one or more **Dense (Fully Connected) layers** to interpret the features and classify the video.

7. Output Layer (Softmax):

- A final Softmax layer outputs the probability distribution over the target classes: **Arrest, Abuse, Arson, Assault, and Background**.
- The class with the highest probability is selected as the prediction.

4.2 3D MAX POOLING

3D Max Pooling is a downsampling operation used in 3D Convolutional Neural Networks. Just like 2D Max Pooling reduces the size of 2D feature maps in CNNs, **3D Max Pooling operates across three dimensions—height, width, and time (depth)—to reduce the spatiotemporal volume of feature maps while retaining the most significant information.**

This operation is essential in crime scene detection tasks where you're analyzing **video clips**, as it helps reduce computational complexity while preserving motion and spatial patterns.

Working of 3D Max Pooling in Crime Scene Detection

- **Input Feature Volume:**
 - After a 3D convolution, the output is a **3D feature map** with dimensions like (frames, height, width) for each channel.
 - For example: A feature map of shape (8, 28, 28) might represent 8 frames of a downsampled video clip.
 - **Pooling Operation:**
 - A **3D kernel** (e.g., $2 \times 2 \times 2$) slides over the feature map in all three dimensions—**time (frames), height, and width**.
 - At each location, it selects the **maximum value** within that 3D window (cube).
 - This captures the **strongest activation** (most important feature) across both space and time.
 - **Strides:**
 - You typically define a stride (e.g., (2, 2, 2)) that determines how far the kernel jumps each time.
 - This effectively **halves** the dimensions of the input in each axis, reducing the volume by a factor of 8.
 - **Resulting Output:**
 - The output is a **smaller 3D feature map** that retains essential information.
 - For example, an input of shape (8, 28, 28) could be reduced to (4, 14, 14).
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4.3 SOFTMAX CLASSIFIER

The **Softmax Classifier** is a mathematical function commonly used in the final layer of deep learning models for **multi-class classification** tasks. It converts a raw vector of scores (logits) into **probabilities**, allowing the model to predict which class an input belongs to with a confidence score for each class.

Working of the Softmax Classifier

1. **Input (Logits):**
 - The model produces a vector of raw, unbounded outputs (called **logits**) from the final dense (fully connected) layer.
 - Example: [3.2, 1.8, 0.5] for 3 possible classes (e.g., "Arrest", "Abuse", "Arson").
2. **Softmax Function:**
 - Applies the following transformation to convert logits into **normalized probabilities**:
$$\text{Softmax}(z_i) = e^{z_i} / \sum_{j=1}^K e^{z_j}$$
 - Where:
 - Z_i : Logit for class i
 - K : Total number of classes
3. **Output:**
 - A probability distribution over all classes that **sums to 1**.
 - Example: [0.72, 0.20, 0.08] → the model is 72% confident the input is "Arrest."
4. **Prediction:**
 - The class with the **highest probability** is selected as the predicted label.
 - Helps interpret which class the model believes is most likely.

Why Softmax is Important in Crime Scene Detection

- **Handles Multiple Classes:** Essential when detecting more than two crime types (e.g., "Abuse", "Arrest", "Arson", "Assault").
- **Provides Confidence:** Gives not just the prediction, but how confident the model is about it.
- **Enables Interpretability:** Outputs are human-understandable probabilities instead of arbitrary scores.
- **Works with Categorical Crossentropy Loss:** Trains the model to match the true label distribution.

4.4 CROSS ENTROPY LOSS

Cross Entropy Loss is a widely used loss function for **classification problems**, especially in combination with the **Softmax classifier**. It quantifies the difference between the **predicted probability distribution** and the **actual (true) distribution**.

What It Does

- Measures **how well the predicted probabilities match the true labels**.
 - Penalizes incorrect predictions more harshly when the model is confident but wrong.
-

- Drives the model to assign **high probability to the correct class**.

Mathematical Definition

For a **single training example** with true label y and predicted probability vector $y^\wedge$:

$$\text{CrossEntropy}(y, y^\wedge) = - \sum_{i=1}^C y_i \log(y^\wedge_i)$$

Where:

- C : number of classes
- Y_i : 1 if class i is the true label, otherwise 0 (one-hot encoding)
- $y^\wedge_i$: predicted probability for class i

Why It's Ideal for Classification

- **Works with Softmax**: Cross entropy pairs perfectly with the softmax output, punishing low confidence for correct classes.
- **Encourages Probabilistic Confidence**: The better the model estimates the right class with high certainty, the lower the loss.
- **Efficient Gradient Flow**: It's smooth and differentiable—ideal for gradient-based learning (backpropagation).

4.5RANDOM CLIP SAMPLING

Random Clip Sampling is a data preprocessing technique used in video analysis tasks to extract **short, fixed-length segments (clips)** from longer videos for **training 3D CNNs or RNNs**. Instead of processing the entire video, this approach selects a **random sequence of consecutive frames** to reduce computational cost and introduce **temporal variation** in training.

Why It's Used

- **Reduces memory usage**: Entire videos are too large to process all at once.
 - **Improves model generalization**: Randomness acts like temporal augmentation.
 - **Ensures uniform input size**: Required for batch processing in deep learning.
-

How It Works

- Let's say you have a video with $T = 120$ frames, and you want clips of $L = 16$ frames:
 - **Determine Valid Start Range**:
The start frame can be any index s such that:
$$0 \leq s \leq T - L$$
 - **Randomly Select Start Frame**: Choose a random s from that range.
 - **Extract Clip**: Take frames from s to $s + L - 1$.
 - **Feed to Model**: The resulting 16-frame clip is now treated as one training sample.
-

4.6 ADAM OPTIMIZER

Adam (Adaptive Moment Estimation) is an advanced optimization algorithm that combines the benefits of two other popular methods: AdaGrad and RMSProp. In crime scene detection, Adam plays a crucial role in efficiently updating network weights to minimize loss and achieve faster convergence.

Working of Adam Optimizer in Crime scene detection

1. Initialization of Parameters

- Adam maintains two moving averages for each parameter:
 - First Moment (Mean):** Tracks the average of gradients (like momentum).
 - Second Moment (Uncentered Variance):** Tracks the average of squared gradients.

2. Gradient Computation

- During each training step, gradients are calculated with respect to the current batch of images (real or fake).
- These gradients represent the direction in which weights should be adjusted to reduce prediction error.

3. Updating Moving Averages

- The **first moment estimate** (m) is updated to smooth out the noise in the gradients.
- The **second moment estimate** (v) captures the scale of the gradients to adapt learning rates individually for each parameter.

4. Bias Correction

- Since initial moving averages are biased towards zero, Adam corrects this bias to ensure accurate updates, especially early in training.

5. Weight Update Step

- Parameters (weights) are updated using both the corrected mean and variance, balancing speed and stability.
- The learning rate dynamically adjusts for each parameter based on past gradients, allowing **faster convergence** without overshooting the minimum.

6. Role in Crime scene detection

- Stabilizes Training:** Helps handle noisy and sparse gradients that arise when detecting subtle manipulations in fake images.
 - Faster Convergence:** Reduces the number of epochs needed for achieving high accuracy.
 - Adaptiveness:** Different weights get different learning rates, making the model more resilient to tricky features in deepfake images.
 - Less Hyperparameter Tuning:** Works well with default settings (learning rate = 0.001), saving experimentation time.
-

5. REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

For implementing **AI-based Crime scene detection with Speech Output**, a system requires sufficient computational power to **process images, extract features, generate captions, and convert text to speech efficiently**. Since deep learning models, such as **CNN, LSTM, and Transformer-based architectures**, require significant computational resources, the hardware should be optimized for handling **image processing, model training, and inference**.

Minimum Hardware Requirements

- ✓ **Processor:** Intel Core i5 (10th Gen or above) / AMD Ryzen 5 equivalent
- ✓ **RAM:** 8GB or higher (16GB recommended for deep learning tasks)
- ✓ **Storage:** 500GB SSD / 1TB HDD (SSD preferred for faster data access)
- ✓ **GPU:** Integrated Intel Iris Graphics (for basic model inference) or NVIDIA RTX 3060+ (for training large models)
- ✓ **Operating System:** Windows 10/11 (64-bit) / Linux (Ubuntu 20.04+)
- ✓ **Software Requirements:** Python, TensorFlow/PyTorch, CUDA (for GPU acceleration), OpenCV, NLTK, and TTS libraries

5.2 SOFTWARE REQUIREMENTS

To implement an **AI-powered crime scene detection system** using video inputs, various software components are required for **video preprocessing, deep learning model training, and real-time inference**. Below is a detailed and categorized list of the essential tools and libraries.

1. Operating System

-  **Windows 11** (*User's current system – suitable for development and testing*)
-  **Ubuntu 20.04+** (*Recommended for GPU support, better performance in training deep learning models*)
-  **macOS Monterey or later** (*Optional – works but GPU acceleration may be limited*)

2. Programming Language

- **Python 3.8+** – Industry standard for AI/ML projects due to its rich ecosystem

3. Development Environment & Tools

- **PyCharm** – Preferred IDE (already used by the user)
 - **Jupyter Notebook** – For visual experimentation, training logs, and data exploration
 - **VS Code** – Lightweight and customizable ID
-

4. Deep Learning & Machine Learning Frameworks

- **PyTorch** – Primary framework for building and training 3D CNNs
- **TensorFlow 2.x** (*Optional alternative*) – For model experimentation or other architecture support
- **Keras** (*High-level API for TensorFlow, not needed if using PyTorch primarily*)

5. Video & Image Processing Libraries

-  **OpenCV** – Core library for video frame extraction, preprocessing, and augmentation
-  **Pillow (PIL)** – Lightweight library for image manipulation

6. Model Architecture & Feature Extraction

-  **Torchvision** – For loading pre-trained models like EfficientNet, ResNet
-  **Pre-trained Models (optional):**
 - EfficientNetB0
 - ResNet18/34
 - Custom 3D CNN (for video clip classification)

7. Data Manipulation & Utilities

-  **NumPy** – Matrix operations and numerical computation
-  **Pandas** – Dataset handling, statistics, and logs
-  **Scikit-learn** – Evaluation metrics like confusion matrix, precision, recall

8. Visualization Tools

-  **Matplotlib** – For plotting accuracy/loss graphs
-  **Seaborn** – Advanced visualization (heatmaps, class distributions)

9. Model Deployment Tools

-  **Streamlit** – For creating a user-friendly web interface to upload and analyze video files
-  **Flask / FastAPI** (*Optional*) – Lightweight REST APIs for backend inference

10. GPU Acceleration & Performance Boost

-  **CUDA Toolkit** – Required if using NVIDIA GPU for training
-  **cuDNN** – Accelerates deep learning computations in PyTorch/TensorFlow
-  **Google Colab / Kaggle Kernels** – For free cloud GPU access (ideal for prototyping and training)

5.3 LIBRARY FILE REQUIREMENTS

To implement **AI-based Crime scene detection**, several Python libraries are required for deep learning, image processing.

These libraries provide pre-built functions and models, making the development process efficient and optimized.

5.3.1 OPENCV

OpenCV is an open-source computer vision and image processing library widely used in AI, robotics, and video analytics projects. It provides a robust collection of tools for **video analysis, frame extraction, object detection, and image transformation**, making it essential for **crime scene detection using video inputs**.

Key Features:

- **Image and Video Processing**
 - Supports reading, writing, and transforming images and videos in multiple formats (e.g., MP4, AVI, JPG, PNG).
 - Ideal for extracting and preprocessing frames from video footage for model input.
- **Frame Sampling & Manipulation**
 - Enables frame-by-frame extraction from videos for creating time-based 3D CNN input clips.
 - Supports resizing, cropping, rotating, and color conversion (RGB ↔ Grayscale).
- **Data Augmentation and Preprocessing**
 - Built-in methods for flipping, blurring, normalization, brightness/contrast adjustment, etc.
 - Useful for augmenting datasets to prevent overfitting in training deep learning models.
- **Object Detection & Image Segmentation (Optional)**
 - Compatible with Haar cascades, HOG descriptors, and DNN modules for lightweight object detection (if needed as a preprocessing step).
- **Integration with Deep Learning Frameworks**
 - Seamlessly integrates with TensorFlow, Keras, and PyTorch by converting images to NumPy arrays for model input.
 - Can be used in real-time inference pipelines with camera feeds or uploaded video files.
- **Platform & Hardware Support**
 - Cross-platform (Windows, Linux, macOS) with support for GPU acceleration via OpenCV CUDA (optional).
 - Lightweight and efficient, even on low-resource devices.

5.3.2 NUMPY

Importance of NumPy in Crime scene detection

- **Efficient Data Handling** – NumPy provides **multi-dimensional arrays (ndarrays)**, which are crucial for storing and processing image data and model parameters.
-

-
- **Matrix Operations** – Neural networks involve matrix calculations (e.g., weight updates, dot products), and NumPy offers **optimized mathematical functions** for these operations.
 - **Image Processing** – Used for **reshaping, normalizing, and transforming** image data before feeding it into deep learning models.
 - **Interoperability** – Works seamlessly with **TensorFlow, Keras, PyTorch, OpenCV, and Pandas** for machine learning tasks.
 - **Vectorized Computation** – Speeds up mathematical operations using **vectorized functions**, reducing execution time compared to standard Python loops.

5.3.3 PYTORCH

PyTorch is a flexible, open-source deep learning framework developed by Facebook's AI Research lab. It's known for its dynamic computation graph and intuitive coding style, making it highly suitable for rapid prototyping and video-based deep learning models like **3D CNNs** used in crime scene detection.

Key Benefits:

- **Dynamic Computation Graph**
 - PyTorch allows on-the-fly model changes, making it ideal for experimentation with custom architectures such as 3D CNNs or hybrid CNN-LSTM models for temporal crime activity analysis.
- **3D CNN Support**
 - Built-in layers like `nn.Conv3d`, `nn.MaxPool3d`, and `nn.BatchNorm3d` enable efficient modeling of spatiotemporal features in video clips (used to detect activities like arson, assault, etc.).
- **Flexible Model Building**
 - Offers low-level control and transparency, which helps in customizing complex models for tasks like frame-wise action classification and video understanding.
- **GPU Acceleration**
 - Provides native CUDA support for training deep models on NVIDIA GPUs, accelerating performance on large video datasets.
- **Seamless Data Handling with Datasets and Dataloaders**
 - Custom Dataset and DataLoader classes help manage large-scale video datasets, random clip sampling, batch processing, and augmentations—all crucial for training temporal models.

5.3.4 TORCHVISION

TorchVision is a specialized computer vision library that comes bundled with PyTorch. It provides a powerful toolkit for image and video processing—making it especially valuable for building models that detect and classify crime scenes from visual data.

Key Benefits:

- **Pre-loaded Datasets for Training & Testing**

- Offers access to popular datasets like ImageNet, CIFAR, and COCO, which can be used directly or adapted for transfer learning or data preprocessing pipelines.
- **Transforms for Image & Video Augmentation**
 - Includes a rich collection of `transforms` (like resizing, cropping, normalization, flipping, etc.) for image and frame augmentation, which improves model generalization and robustness.
- **Video Frame Processing**
 - With utilities like `torchvision.io.read_video`, TorchVision helps extract frames directly from videos, enabling clip-based training for 3D CNNs used in detecting temporal patterns like arson or assault.
- **Pre-trained CNN Models**
 - Provides quick access to pre-trained deep learning models (ResNet, VGG, Inception, EfficientNet, etc.) used for image feature extraction—essential in crime classification pipelines.
 - These models can be used for **transfer learning**, drastically reducing training time and improving accuracy.
- **Efficient Image Loading & Batching**
 - Works hand-in-hand with PyTorch's `DataLoader` and `Dataset` for loading and batching large image/video datasets efficiently.
- **Smooth Integration with PyTorch**
 - TorchVision models and transforms are natively compatible with PyTorch tensors and optimizers, making the full pipeline cohesive and easy to manage.
- **Utility Functions for Visual Analysis**
 - Functions like `make_grid()` and `utils.draw_bounding_boxes()` support visualization of model predictions, bounding boxes, and detection outputs—important for interpreting crime detection results.

5.3.5 TQDM

Importance of tqdm in Crime Scene Detection

TQDM is a lightweight Python library used to display smart progress bars in loops, data processing tasks, and model training workflows. It provides real-time feedback and transparency, making long-running tasks more manageable and trackable.

Key Benefits:

- **Real-Time Progress Monitoring**
 - Displays dynamic progress bars while loading datasets, preprocessing frames, or training models—giving you visibility into how much of the task is complete and how long remains.
- **Debugging Aid**
 - Helps identify bottlenecks by visually showing where in the data pipeline or training process things are slowing down.

- **Model Training Insight**
 - When training CNN or 3D CNN models over multiple epochs, `tqdm` shows the progress of each batch, helping monitor long training sessions with ease.
- **Loop Transparency in Evaluation**
 - Useful during validation/testing loops over large datasets, ensuring smooth operation and easy interruption handling if needed.
- **Seamless Integration with PyTorch and Keras**
 - Works effortlessly with `DataLoader`, `enumerate()`, `range()`, and even model training loops in PyTorch/Keras—requiring minimal code changes.
- **Real-Time ETA and Performance Metrics**
 - Provides Estimated Time of Arrival (ETA) and items per second metrics, aiding in time management and workflow planning.
- **Customizable & Lightweight**
 - Can be integrated into any script with just one line, and supports nested loops, custom descriptions, and post-loop summaries with negligible overhead.

5.3.6 OS

Importance of os in Crime Scene Detection

The `os` module in Python is a standard library used for interacting with the operating system. In AI-based crime scene detection systems, it plays a critical role in managing datasets, organizing files, and ensuring smooth file path operations across platforms.

Key Uses:

- **Dataset Directory Management**
 - Navigates through folders containing categorized video/image datasets (e.g., arrest, arson, abuse, assault) to load them for training or testing.
- **File Path Handling**
 - Constructs cross-platform-compatible paths (using `os.path.join()`), reducing bugs when switching between Windows, Linux, and macOS environments.
- **Automated Cleanup and Organization**
 - Helps remove temporary files or automatically organize augmented/processed images into appropriate folders.
- **Batch Processing**
 - Iterates over directories to apply operations (e.g., renaming files, loading frames from video folders) to all dataset items in a loop.
- **Model Checkpointing and Logging**
 - Dynamically creates folders to save model checkpoints, training logs, and generated outputs like captions or prediction results.
- **Automation of Training Pipelines**
 - Enables writing scripts that detect and adapt to directory structures, making your training pipeline robust and easily transferable.
- **Cross-Platform Portability**
 - Ensures your project works across development, testing, and deployment environments without path issues.

5.3.7 STREAMLIT

Importance of Streamlit in Crime scene detection

- **Rapid Prototyping** – Enables the creation of interactive front-end applications with just Python, allowing developers to deploy deep learning models quickly without needing web development skills.
- **User Interface for Image Upload** – Provides easy-to-use widgets like `file_uploader()` to let users upload images, essential for testing crime scene detection in real-time.
- **Real-Time Feedback** – Displays model predictions instantly using `st.success()`, `st.error()`, or `st.image()`, offering a responsive user experience for end-users.
- **Integration with ML Models** – Seamlessly integrates with TensorFlow/Keras models via simple Python code. It allows predictions to be run live as users interact with the app.
- **Caching Mechanism** – Uses decorators like `@st.cache_resource` to avoid reloading the model every time the user interacts, improving performance and load time.
- **Cross-Platform Deployment** – Streamlit apps can be deployed locally, on cloud servers, or shared publicly with one command (`streamlit run`), making your deepfake detector easily accessible.

6. IMPLEMENTATION

Step 1 – Environment Setup

Library	Purpose in this Project
PyTorch	Defining, training, and evaluating the 3D CNN model
OpenCV	Extracting video frames and converting them into tensors
NumPy	Frame indexing and manipulation
Streamlit	Interactive GUI for uploading crime videos and showing results
Torchvision	Image transformations (optional, extendable)
tqdm	Visualizing training progress

To install all required libraries:

```
$  
pip install torch torchvision opencv-python numpy streamlit tqdm\  
$
```

Step 2 – Model Training (training.py)

1. Data Loading & Preprocessing

Each video is loaded and a random 16-frame clip is extracted. Frames are resized to 112x112 pixels and converted from BGR to RGB (since OpenCV loads in BGR by default). The pixel values are normalized between 0 and 1 by dividing by 255.

```
$  
frame = cv2.resize(frame, (IMG_SIZE, IMG_SIZE))  
frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)  
frame = torch.tensor(frame, dtype=torch.float32).permute(2, 0, 1) / 255.0  
$
```

Key points:

- Ensures uniform input size for the CNN.
- Maintains consistent color channels.
- Efficient tensor stacking for 3D CNN input.

The dataset is dynamically generated using a custom CrimeVideoDataset class:

- Video files are read from folders named after crime classes (dataset/Arson, dataset/Assault, etc.).
- Labels are inferred from folder names.

2. CNN Architecture – 3D Convolutional Network

The model uses 3D convolution to extract **spatiotemporal features** from video clips.

```
$  
nn.Conv3d(3, 32, kernel_size=3, stride=1, padding=1)  
$
```

Network Highlights:

- **3D Conv Layers:** Capture both spatial and temporal patterns.
- **MaxPooling3D:** Downsamples features in time and space.
- **Linear Layers:** Learn to classify high-level features into specific crime categories.
- **Dropout Layer:** Prevents overfitting.

Final tensor shape calculation:

```
$  
64 channels × 4 frames × 28 × 28 pixels → flattened for dense layer  
$
```

3. Compilation & Training

The model is trained using:

- **CrossEntropyLoss:** Ideal for multi-class classification
- **Adam Optimizer:** Fast convergence and adaptive learning

```
$  
criterion = nn.CrossEntropyLoss()  
optimizer = optim.Adam(model.parameters(), lr=0.001)  
$
```

Training runs for 10 epochs with batch size 2. Model performance is shown live using tqdm progress bars.

4. Model Saving

The trained weights are saved as a PyTorch checkpoint:

```
$  
torch.save(model.state_dict(), "crime_model.pth")  
$
```

Step 3 – Web-based GUI using Streamlit (main.py)

1. Model Loading

On app load, the model is reconstructed and pre-trained weights are loaded.

```
$  
model.load_state_dict(torch.load(MODEL_PATH, map_location=DEVICE))  
model.eval()  
$
```

\$

This allows for quick, GPU/CPU-compatible predictions.

2. Video Upload & Clip Extraction

User uploads a video clip (.mp4/.avi). The app:

- Saves it temporarily
- Picks a random 16-frame clip
- Processes each frame just like in training

\$

```
video_tensor = video_tensor.permute(1, 0, 2, 3) # (3, 16, 112, 112)
video_tensor = video_tensor.unsqueeze(0).to(DEVICE)
```

\$

Ensures the input shape matches model expectations.

3. Prediction Logic

The model outputs class logits. The prediction is made using:

\$

```
pred_idx = torch.argmax(probs, dim=1).item()
```

\$

The predicted crime class is shown along with emoji-based success feedback.

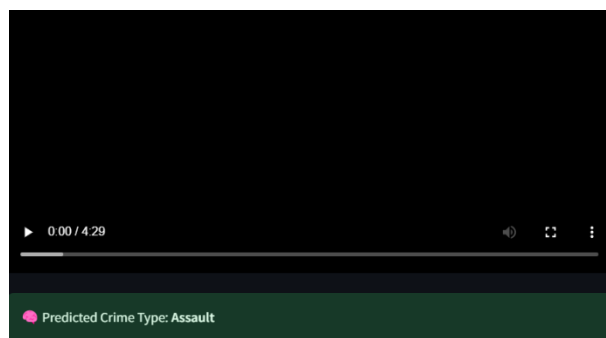
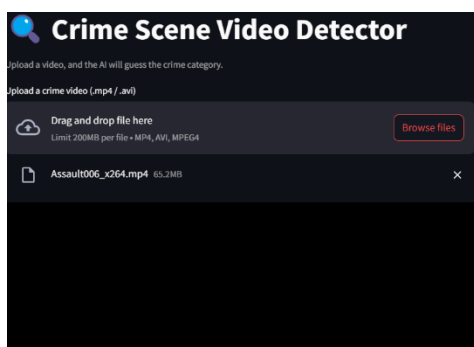
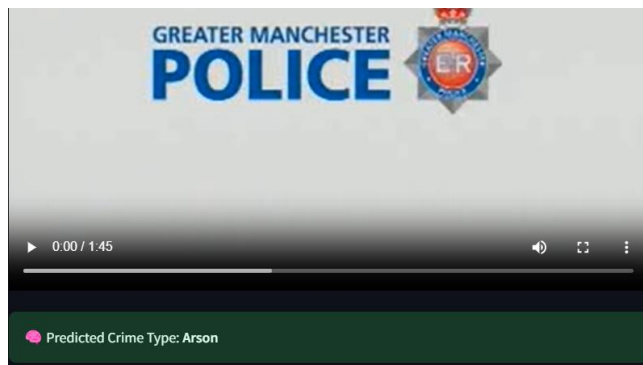
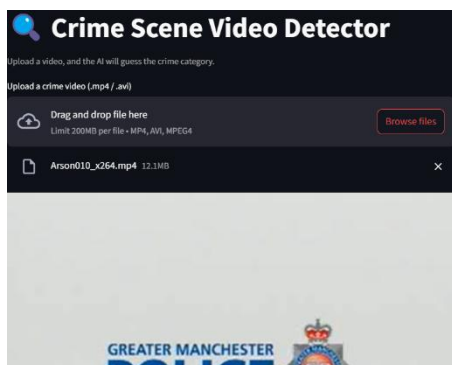
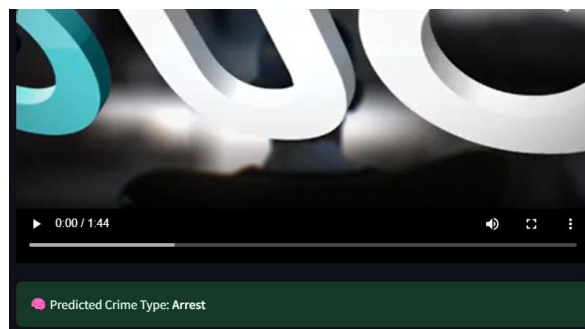
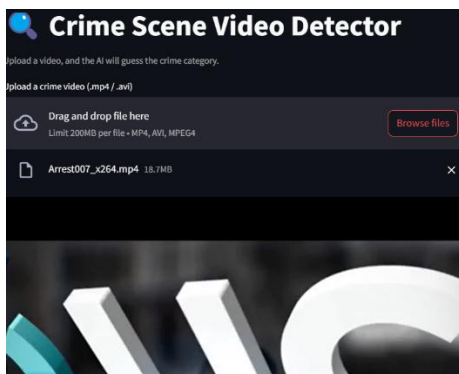
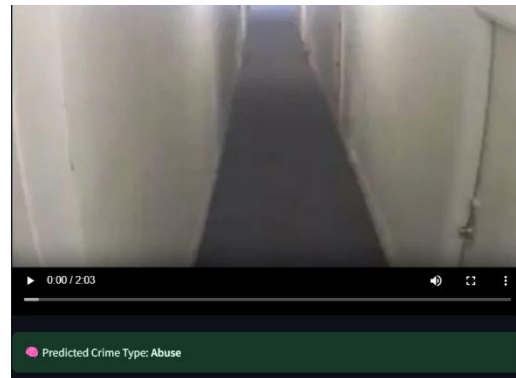
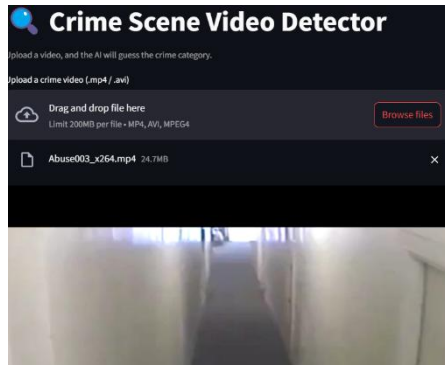
End-to-End Execution Flow

1. User uploads a crime video through Streamlit.
2. A 16-frame clip is randomly extracted and preprocessed.
3. The trained 3D CNN model is loaded and run on the clip.
4. Predicted crime class is shown as result (e.g., **Assault**, **Arrest**, etc.).

✅ Why This Implementation Works

- **3D CNN:** Learns both spatial and temporal patterns—key for video-based detection.
- **Custom PyTorch Dataset:** Makes the model scalable to any number of classes or new video inputs.
- **Lightweight yet Effective Architecture:** Balances performance and compute requirements.
- **Streamlit:** Allows real-time inference and interactive feedback in just a browser.

7. RESULTS



8. CONCLUSION & FUTURE SCOPE

The implementation of the **AI-based Crime Scene Detection System** demonstrates the power of deep learning in enhancing public safety through automated video analysis. Leveraging a **custom 3D Convolutional Neural Network (3D CNN)**, the model efficiently processes temporal and spatial features from short video clips to classify them into four critical categories: **Arrest, Abuse, Arson, and Assault**.

The use of a **Streamlit frontend** provides a clean and user-friendly interface, allowing users to upload videos and receive instant feedback on the type of crime detected. PyTorch powers the backend for model training and inference, enabling high performance even with modest hardware.

This project showcases how modern computer vision techniques can be used to support law enforcement, surveillance systems, and public safety organizations by automating the initial review of video evidence, thereby reducing manual effort and increasing response time.

Future Scope

While the system is a solid proof of concept, it opens up many exciting possibilities for further research, development, and deployment:

- ✅ **Real-Time Crime Detection in CCTV Feeds** – Integrate with live CCTV systems using OpenCV to analyze and classify crimes in real time, triggering alerts for faster incident response.
- ✅ **Action Localization & Bounding Boxes** – Combine 3D CNNs with action detection models like SlowFast or I3D to not only classify the crime but also highlight the region or time segment where the activity occurs.
- ✅ **Scene and Context Awareness** – Use object detection and background segmentation (e.g., Detectron2, YOLOv8) to capture contextual clues like fire, weapons, or crowd behavior to support multi-modal crime classification.
- ✅ **Multi-Class and Multi-Label Classification** – Expand beyond four classes to detect other crime types such as theft, vandalism, or riots, and support overlapping labels for complex scenes.
- ✅ **Explainable AI (XAI)** – Implement Grad-CAM or LIME to visualize what parts of the video influenced the model's prediction, increasing transparency and user trust.
- ✅ **Voice-Assisted Feedback** – Incorporate TTS (Text-to-Speech) modules that announce the predicted crime type, making the tool accessible for visually impaired users or low-visibility control rooms.

- ✓ **Edge Deployment** – Optimize and deploy the model on edge devices (e.g., NVIDIA Jetson Nano, Raspberry Pi) for on-site surveillance and decentralized detection.
- ✓ **Cloud API Integration** – Offer the detection model as an API through platforms like AWS SageMaker, GCP Vertex AI, or FastAPI, making it easily integrable with third-party systems or law enforcement databases.
- ✓ **Mobile App Companion** – Develop a mobile app for officers or the public to upload suspicious clips and receive crime classification feedback instantly, aiding citizen-led reporting and investigations.
- ✓ **Crowdsourced Dataset Expansion** – Allow users to submit new clips with labels, creating a feedback loop for continual model training and improvement with real-world data.

This project lays the groundwork for a next-generation crime detection platform that blends AI with civic technology. With further development, it can play a vital role in **smart city infrastructure, law enforcement assistance, and crime prevention initiatives**.

Final Thoughts

This project exemplifies the transformative power of artificial intelligence in digital media authentication. By leveraging **deep learning, transfer learning, and intuitive deployment frameworks**, the system not only detects deepfakes with high precision but also serves as a frontline defense against visual misinformation. The combination of EfficientNet’s efficiency and Streamlit’s simplicity showcases how complex AI models can be made accessible to end-users, bridging the gap between cutting-edge research and real-world usability.

At its core, this project illustrates the evolution of machines from mere data processors to intelligent visual auditors capable of making informed decisions based on visual content. Just as humans use context and intuition to judge the authenticity of media, AI models like this are beginning to emulate those cognitive processes—detecting manipulation patterns that escape the naked eye. As we continue to improve model accuracy, training data diversity, and deployment strategies, the system becomes increasingly adept at understanding not just *what* it sees, but *why* it matters.

Furthermore, this technology holds immense promise beyond academic or prototype environments. In **media houses**, it could validate image authenticity before publication. In **law enforcement**, it could assist in forensic investigations involving doctored evidence. In **finance and governance**, it could help detect altered IDs or documents submitted for verification. And in **education**, it can serve as a tool for teaching digital literacy and responsible AI practices.

As AI continues to mature, the fusion of **visual intelligence, interpretability, and real-time deployment** will define the next generation of intelligent systems. Future innovations could include **integrated AR/VR overlays to flag fake content in real time, automated policy enforcement on social platforms, and hybrid models combining vision with audio and text forgery detection**—ultimately leading to a safer, more trustworthy digital landscape.

This project is not just a proof of concept; it’s a glimpse into a future where machines actively safeguard truth in the visual age.

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